Exercise 3.6

1. Express the first quantity as a fraction of the second quantity, and give each fraction in its lowest terms:

(i) 16, 24 (ii) 3 days, 1 week (iii) 40 c, €2

(iv) 24 mins, 2 hours (v) €1.20, €3 (vi) 8 hours, 3 days

2. Hannah spends $\frac{1}{5}$ of her pocket money on sweets and $\frac{2}{3}$ on clothes.
She saves the rest.
What fraction of her pocket money does Hannah save?

3. The first $\frac{1}{2}$ of TANK is TA. The last $\frac{1}{3}$ of IRRITABLE is BLE.
Put them together and you get TABLE.
What words do these make?

(i) The first $\frac{1}{3}$ of TRIANGLES and the second $\frac{1}{2}$ of STICKS

(ii) The first $\frac{2}{5}$ of PRECIPICES and the last $\frac{2}{5}$ of VICTORIOUS

(iii) The first $\frac{1}{3}$ of DISCUSSED, the first $\frac{1}{4}$ of APPLICATIONS and the last $\frac{3}{4}$ of FEAR

4. A journey to school consists of a $\frac{2}{5}$ km walk, then a $3\frac{1}{4}$ km bus journey, and finally a $\frac{1}{2}$ km walk from the bus stop to the school.
Find the total length of the journey to school.

5. Find $\frac{3}{8}$ of €576.

6. In a bag of 56 coloured counters, $\frac{1}{4}$ are red, $\frac{2}{7}$ are blue and the rest are yellow.
How many of the counters are yellow?

7. Emer bought a 3-hour blank video tape. She recorded programmes lasting $1\frac{1}{2}$ hours, $\frac{3}{4}$ hour and $\frac{2}{3}$ hour on this tape.
How many minutes recording time were left on the tape?

8. Three quarters of a certain number is 27.

Find (i) $\frac{1}{4}$ of the number.
(ii) the whole number.

9. Alan spent $\frac{4}{9}$ of his money. If he had €65 left, how much had he at first?

10. How many $1\frac{1}{4}$ litre cartons of orange juice can be filled from a container holding 40 litres?

11. $\frac{3}{7}$ of the students in first-year are girls.
If there are 64 boys in first-year, find the number of girls in the year.

12. $\frac{2}{5}$ of the spectators at a football match were in the terraces and the remaining 17 580 were in the stands. Find the total number of spectators at the match.

13. How many $\frac{1}{4}$ litre glasses can be filled from a container which contains $4\frac{1}{2}$ litres?

14. A ferry completes one crossing in $1\frac{1}{2}$ hours.
What is the greatest number of crossings it can make in a 24 hour period?

15. Which of the letters on the line below corresponds to $\frac{5}{8}$?

16. A class of 30 students went to a fast-food bar.
$\frac{3}{10}$ of them bought a hamburger, $\frac{2}{5}$ of them bought a pizza and the remainder bought chips.

(i) What fraction of the students bought chips?

(ii) How many students bought a pizza?